Toward a Unit of AI Inference: Auditing Resource-Ratio Calibration

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Abstract

Resource ratios between tasks offer a tempting way to normalize language-model efficiency across heterogeneous workloads. Such conversions require more than a close pooled fit: their meaning depends on acceptance rules, execution boundaries, reference systems, and the distribution of task comparisons. We audit these requirements in three public evaluation archives using local reanalysis. Across 81,438 matched Reasoning Gym first completions, median omitted-system reconstruction errors for visible response bytes are 2.308 and 2.596 in easy and hard settings; bytes omit separately returned reasoning. Independent exact arithmetic finds incorrect references for 28 of 50 hard decimal-sum questions, producing 117 archived rejections of exact correct answers. In a restricted METR panel of 6,996 runs, aggregate-token reconstruction lies within factor 1.25 for 75.8% of comparisons, with 67.3% coverage under exclusion of the held label's provider proxy; individual-system coverage ranges from 48.5% to 97.4%. An engineering release exposes outcome-dependent missing telemetry and aggregation sensitivity. Constructed studies distinguish taxonomy effects, sparse-bridge identification, and accounting-error bounds from empirical validation. The contribution is a reproducible audit of resource-ratio calibration and its failure modes. None of the archives identifies minimum resource cost at a common reliability target or validates a unit of useful inference work.

1 Introduction

A task-based efficiency comparison needs a defensible account of both the outcome and the resources used to obtain it. Counting tokens records an execution input, while counting accepted completions records an outcome under a specified acceptance rule. Comparing heterogeneous tasks adds a further choice: how should their contributions be weighted? One possible convention derives task weights from resource ratios measured on reference systems. If those ratios are stable across systems, a calibrated task index could support interpretable comparisons. If they are not, an apparently precise scalar may depend strongly on its reference panel or workload.

Our research question is deliberately limited: *what do released evaluation records support about resource-ratio calibration, and which diagnostics reveal when a favorable reconstruction summary is misleading?* This is an evidence audit, not an experiment that measures a new inference unit. A recorded observable can be analyzed consistently without measuring complete inference resources; a correct resource table can still lack valid outcome labels; and a close pooled median can hide poor transfer between domains or systems. These distinctions matter to researchers reusing public evaluation traces for efficiency indexes.

We examine three complementary releases. Reasoning Gym offers broad task-family coverage and stored responses, enabling an exact matching analysis and an independent numeric-answer audit. A METR release provides aggregate-token counts on a narrowly defined simple-task panel, enabling controlled sensitivity

analyses within one recorded observable. A creator-reported engineering archive exposes the effect of missing telemetry and task aggregation. These archives are convenience cases selected for public trace availability. Differences between them do not identify effects of task difficulty, resource quality, or model capability.

The contributions are: (1) an explicit distinction between a recorded-resource index and a reliability-qualified service cost, together with elementary consistency diagnostics; (2) an independently reproducible acceptance audit showing how small reference-value errors change exact grading; and (3) retrospective tests of system-panel, domain, reference, repeat-count, selection, and taxonomy dependence. The new sensitivity analyses were added after inspecting the original results and are labeled accordingly. Constructed graph and accounting studies establish properties of the diagnostics, not properties of deployed models.

The strongest finding is not that every scalar index is impossible. It is that the evaluated pooled summaries provide insufficient evidence for a stable conversion across the relevant groups. A declared reference index can remain useful as a convention, but its task measure, reference panel, acceptance rule, and resource boundary must remain part of its meaning.

2 Related work and contribution boundary

Quality, workload, and efficiency. Joint accuracy–cost evaluation and quality-constrained benchmarking already distinguish useful outcomes from resource consumption (Kapoor et al., 2025; Reddi et al., 2020). OckBench, procedural difficulty studies, and test-time allocation examine correctness, workload, and execution policy (Du et al., 2025; Laber et al., 2026; Snell et al., 2025). Task-denominated prices and quality-adjusted AI price indexes address related economic questions (Gundlach et al., 2026; Zhu, 2026). Kaiser et al. (Kaiser et al., 2026) are the closest AI-efficiency precedent: they study workload normalization, declared or solver-derived references, and reference-pool robustness, including metric comparisons. We do not introduce those ideas. Our distinctive evidence concerns the fitness of released resource and outcome records and the sensitivity of *task-ratio reconstruction* under explicitly fixed comparisons. Zhu (Zhu, 2026) also demonstrates stable rankings alongside a materially changed economic index. Neither stable rankings nor a single pooled error summary establishes stable conversion magnitudes.

Reference normalization and multilateral indexes. SPEC normalizes benchmark results against a fixed reference and aggregates ratios within a declared suite; its documentation warns against conversion between suites (SPEC, n.d.). Country-product-dummy methods regress log prices on country and product effects (Summers, 1973; Asian Development Bank, n.d.), the same model class as system and task effects in log resource. Multilateral price-index research also studies transitivity and spatial linking (Hajargasht et al., 2017). The algebraic model and graph conditions are therefore established tools, not new general measurement results. Our application differs in the outcome-verification, incomplete execution-boundary, and policy-qualification questions raised by LLM traces.

Capability linking and measurement. Item-response and benchmark-linking methods infer capability from overlapping success data (Maia Polo et al., 2024; Ho et al., 2025). Human-time task horizons similarly concern performance at a reliability threshold (Kwa et al., 2025). They do not automatically define additive weights for completed work. Measurement units are conventions with declared comparison rules (Joint Committee for Guides in Metrology, 2012); additive representations and graph consistency have a long history (Luce & Tukey, 1964; Jiang et al., 2011). The reference-solve proposal in Appendix F motivates the audit but is not an empirically validated contribution. Routing is also separate: established accuracy–cost methods (Chen et al., 2023; Ong et al., 2024) do not require a common work unit, and a fixed request weight cancels from a same-request model-choice ratio.

3 Audit objects and diagnostics

3.1 Recorded means and reconstruction

Let $Z_{mi} > 0$ be the arithmetic mean of a declared recorded observable for system m on task or family i. The observation unit, repeated-run policy, failures, and task aggregation are specified separately for each archive.

The scalar restriction is $Z_{mi} = s_m w_i$, with positive system factors s_m and task weights w_i. On a complete matrix, fitting system and task effects by equally weighted squared error in $\log Z$ gives task differences from column means.

For each omitted system m, estimate task effects from the other systems and observe only that system's anchor-task mean Z_{ma}:

$$\widehat{\beta}_i^{(-m)} = \frac{1}{M-1} \sum_{n \neq m} \log Z_{ni}, \qquad \widehat{Z}_{mi}^{(a)} = Z_{ma} \exp(\widehat{\beta}_i^{(-m)} - \widehat{\beta}_a^{(-m)}). \tag{1}$$

We report $F = \exp|\log(\widehat{Z}/Z)|$, its median and 90th percentile, and the fraction $B_{1.25} = \Pr(F \leq 1.25)$ under the declared empirical comparison measure. Quantiles are computed on absolute log error and exponentiated. The band is descriptive, not an established equivalence criterion. Every distinct anchor/target pair is included. Reversing a pair duplicates its absolute error; pairs and folds share data and are not independent replications. Observing an anchor mean requires repeated execution, not one API request. The targets are the same released task identities, so this is retrospective reconstruction rather than future-instance prediction.

The original summary weights system labels and ordered distinct task pairs equally. This is a *category-based empirical measure*, not a deployment workload. We retain it to reproduce the initial analysis and report alternatives instead of replacing it after seeing results. A measure over underlying tasks and pairs must be fixed before a taxonomy-independent interpretation is possible (Section 7).

3.2 Controls, comparisons, and uncertainty target

Constant weights predict the anchor mean for every target. A reference-system control uses that system's task ratios. The original control chooses the first alphabetically ordered nonomitted label; new sensitivity analyses consider every preferred reference, falling back to the first eligible label in its own omitted fold. All such rules use identical held folds and targets. Reference choice is not optimized on the reported evaluation set. Prompt and gold-answer byte controls are available only in Reasoning Gym; gold-answer length is an oracle diagnostic, not an available deployment feature.

The primary estimand is a deterministic summary of the *recorded cell means*. It needs no sampling confidence interval. Repeated-run resampling, half-repeat panels, and a rank-one noise simulation are separate diagnostics for the estimation procedure. They are not confidence intervals for future systems, unseen instances, or qualified cost. Model-dependent noise simulations cannot distinguish all real interaction from estimation error. We retain their assumptions and compare like summaries rather than advertising displaced percentile ranges as inferential uncertainty.

3.3 Three evidence requirements

Table 1 separates three questions that a scalar fit cannot answer. Traceability supports reproducible arithmetic; it does not establish that an observable is a complete resource or that recorded acceptance matches the intended outcome.

4 A broad response archive and an acceptance audit

4.1 Source, selection, and resource boundary

Reasoning Gym supplies procedural tasks with verifiable rewards (Stojanovski et al., 2025). We inspect all 2,049 released zero-shot files at a pinned evaluation-archive commit (Open Thoughts, 2025). One hard run declares a different model in its summary than in its directory; it is excluded rather than assigned by guesswork. Within the remaining ten easy and nine hard system records, we match exact question and archived expected-answer strings, retain the first duplicate, require common system prompts, at least 20 shared unique instances per family, and positive mean bytes. The resulting panels contain 88 and 86 families and 43,260 plus 38,178 first-completion records. Matching the reference string controls source-version mismatch but does not certify the reference.

Table 1: Recorded quantities and unresolved evidence. None of these releases evaluates a policy-menu minimum at an independently qualified common reliability target. Cross-row comparisons change several factors and have no causal interpretation.

Archive	Recorded quantity	Outcome evidence	Scope restriction
Reasoning Gym	First stored response bytes; separate reasoning and native usage absent	Source scores; six numeric grammars audited independently	Exact matched instances, separate easy/hard panels
METR simple tasks	Reported aggregate tokens; native components and monetary cost unavailable	Source acceptance; sparse repeats per exact task	One named scaffold and task version; 66 short task identities
Engineering	Native usage, scheduled-price estimate, elapsed time	Creator-reported outcomes; outputs and private graders unavailable	Missing usage associated with failure; matched subsets change task coverage

Historical evaluator code retains `message.content`, not complete native usage or separately returned reasoning. The observable is therefore UTF-8 bytes of the first stored response, including empty strings and failures. We do not select successful or best-of-many outputs. Empty stored content and unavailable transport failures are different states; only the former is observed. The resulting reconstruction diagnoses the visible channel, not computation or inference efficiency.

Table 2: All-anchor omitted-system reconstruction of stored visible bytes. Medians are multiplicative error factors. Comparisons are dependent; these are finite-panel summaries, not inference-cost estimates. Full-panel bootstrap intervals are unavailable owing to zero-mean resamples.

Weight rule	Easy median	Hard median	Easy $F \leq 1.25$	Hard $F \leq 1.25$
Shared scale	2.308	2.596	15.0%	13.8%
Constant weights	2.361	2.741	17.2%	14.6%
Prompt bytes	3.021	4.170	11.4%	8.5%
Gold-answer bytes (oracle)	4.689	6.087	8.1%	7.4%
One reference system	2.280	2.570	15.2%	14.2%

Median factors are 2.308 and 2.596, with only 15.0% and 13.8% within 1.25. A single reference has slightly smaller median errors. Whole-family exclusion of empty first responses still yields median factors 2.089 and 2.028; the post hoc restriction does not restore omitted reasoning. Figure 1 illustrates one mechanically chosen anchor, while all numerical summaries include every eligible anchor.

4.2 Independent arithmetic and scoring consequences

We parse six released numeric grammars: integer and decimal chain sums, products, greatest common divisors, least common multiples, and bit counts. These were chosen for inspectable printed operands and an independently implementable exact parser, not as a random sample of benchmark graders. The audit covers 5,700 source records and 597 distinct questions. It imports no source grader. Decimal sums are crosschecked with rational arithmetic. Extraction uses the last complete answer tag and documented numeric parsing; format-policy disagreement remains separate from arithmetic error.

Twenty-eight of 50 distinct hard decimal-chain references disagree with exact sums of their printed operands; the other 547 distinct audited questions agree. Across repeated model records, 117 archive rejections concern exact correct answers on the affected decimal questions. Two additional rejections concern comma formatting. There are also two archive acceptances unequal to the exact requested sum. These are finite-record counts, not prevalence estimates for unaudited grammars or current releases.

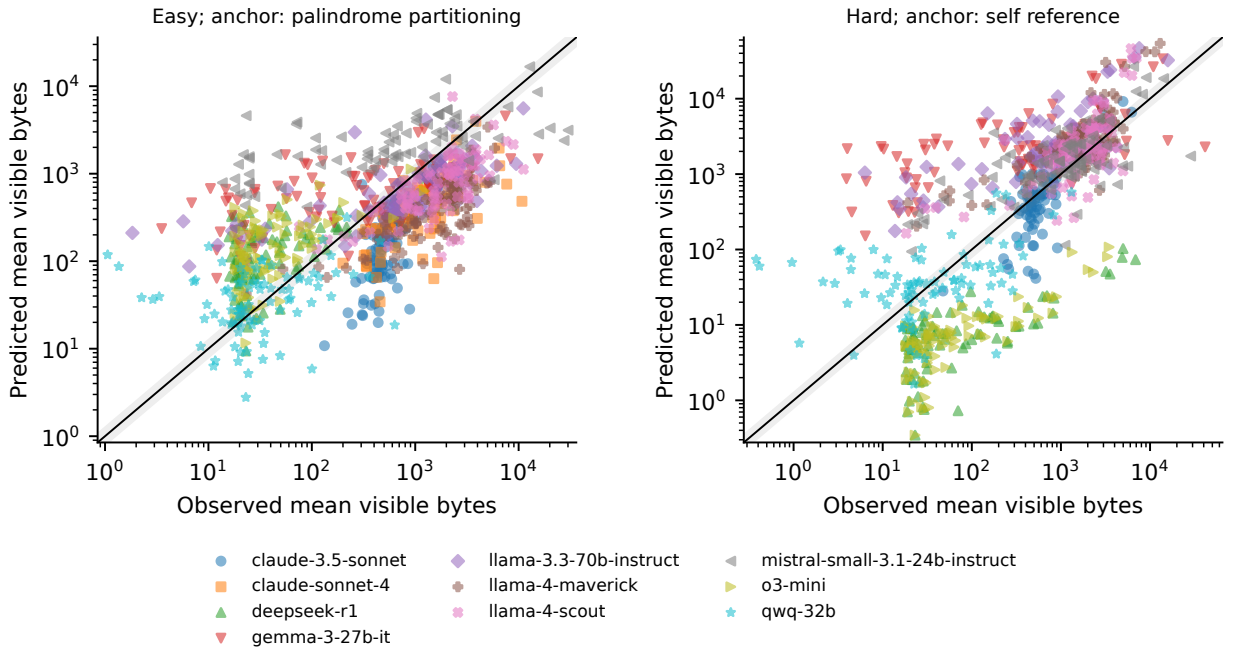


Figure 1: Mean stored response bytes reconstructed for omitted systems using one mechanically chosen anchor per setting. The line marks equality and shading marks factor 1.25. Numerical summaries use all anchors. Visible-byte disagreement does not establish disagreement in unobserved complete inference costs.

Table 3: Examples from distinct hard decimal questions, selected for minimum absolute, maximum absolute and maximum relative reference discrepancy. The released CSV gives hashes, exact operands and all 100 audited decimal questions. Exact grading can reject a mathematically correct answer despite a small relative reference error.

Printed expression	Stored reference	Exact sum
`9398.81320 + 8927.7343 - 7078.67401 + 7665.11455 - 1557.348769 - 3298.2136 - 3607.3605`	`10450.06517`	`10450.065171`
`24895118.82000 + 91430752.870000 + 87815839.73000 + 42664059.2700 - 67964249.460000 - 72979237.160000 - 45633810.2500 + 54211520.3900`	`114439994.1`	`114439994.21`
`555698.34150 - 358211.591300 + 925180.9561 - 567335.234200 - 408324.19110`	`147008.2807`	`147008.281`

The incorrect references differ from exact sums by 0.000001 to 0.11 in absolute magnitude; the largest relative discrepancy is 2.04×10^{-9}. Tiny relative errors matter because the inspected scorer compares Decimal values exactly. For hard sums, DeepSeek R1 changes from 22/50 archive credits to 50/50 exact extracted answers; Claude 3.5 Sonnet changes from 21 to 45 and o3-mini from 20 to 46. This is an acceptance-boundary failure, not evidence that all reference values or all graders are unreliable.

Historical source inspection reveals a compatible mechanism: another generator mutates the Decimal context, and chain sums use inherited precision. Recomputing the 50 hard questions at precision ten reproduces all archived references. This supports a reproduced mechanism without revealing every original runtime event. The pinned-archive rate is not a current-release error rate, and we do not claim first discovery. The original scores remain preserved; corrections are restricted to the audited grammars.

Table 4: METR robustness on identical 18 omitted labels, 66 tasks and all ordered distinct task pairs. Every method observes the same one held-system anchor mean. Excluding provider groups also changes pool size and composition. Labels and pairs are dependent, not independent sample units.

Calibration rule	Median F	90th F	$B_{1.25}$
Equal-label pooled scalar	1.119	1.424	75.8%
Constant weights	1.410	2.281	37.8%
Provider-balanced pool	1.147	1.523	69.2%
Provider proxy excluded	1.149	1.538	67.3%
Excluded, remaining groups balanced	1.178	1.659	61.4%
Median task-effect control	1.123	1.466	74.8%

5 Robustness within one aggregate-token panel

5.1 Fixed panel and recorded outcomes

From METR's public 24,008-run release (Kwa et al., 2025; METR, 2026), metadata selection fixes scaffold `metr_tasks_swaa`, entry point `init_solver,generate`, and task version 2.0.1 before ratio analysis. This yields 6,996 runs, 18 released system labels and 66 common tasks: 20 arithmetic, 20 file-selection, ten alert-triage, nine request-routing and seven code-completion identities. Cells contain four or six runs. These are short simple tasks, not the broad long-horizon workload associated with METR's headline horizons.

Every selected run has a unique identity, positive `tokens_count`, no clone flag and no fatal-error flag. The 44 scored failures remain in the means. The source omits input/output/cache/reasoning components, so we call the observable METR-reported aggregate tokens. All selected `generation_cost` values are zero and are treated as unavailable monetary evidence. Null time limits and lifecycle timestamps do not support a pure latency analysis. The scaffold name and released system label do not establish immutable model snapshots or every policy parameter.

The source reports 6,952/6,996 successes, but exact-task cells include 0/6. A model-family cell has 48/54 successes; 31 of 44 failures occur on `request_routing_9`. Concentration motivates inspection, not a conclusion that its grader is wrong. Four or six runs on one task identity cannot qualify a population reliability requirement. An all-observed-success restriction is an outcome-selected sensitivity, not a replacement calibration sample.

5.2 System panel, domain and reference dependence

The original pooled scalar has median factor 1.119 and $B_{1.25} = 75.8\%$. Individual held-system coverage ranges from 48.5% to 97.4%. Same-family comparisons cover 89.3%, cross-family comparisons 71.9%, and arithmetic/request-routing pairs 55.6% (Figure 2). These comparisons use the same resource matrix: they expose heterogeneity rather than explain a causal mechanism.

The 18 labels map provisionally to eight Claude releases, nine OpenAI releases and one Gemini release; the mapping includes the source's GPT-5.3-Codex alias for `flamingo_2`. Excluding the held label's provider group lowers coverage to 67.3% and raises median factor to 1.149. This also changes calibration-pool size and composition, and is not a causal provider effect or a target-leakage diagnosis. Provider-balanced pooling supplies a further convention sensitivity in the supplement. Only one Google label is available, limiting provider comparisons.

For all 18 preferred-reference rules on identical held folds, median factors range from 1.096 to 1.309; 9 have lower medians than pooling. The range is not a license to select the best reference retrospectively. It shows why an alphabetical baseline cannot establish a general pooling advantage. A versioned index should fix its reference rule independently of its evaluation results.

The largest recorded rectangle forces at least one scalar cell error of 2.058 (Appendix B). Its cells have unequal observed acceptance, so it is not equal-quality specialization. Neither this extreme nor favorable medians

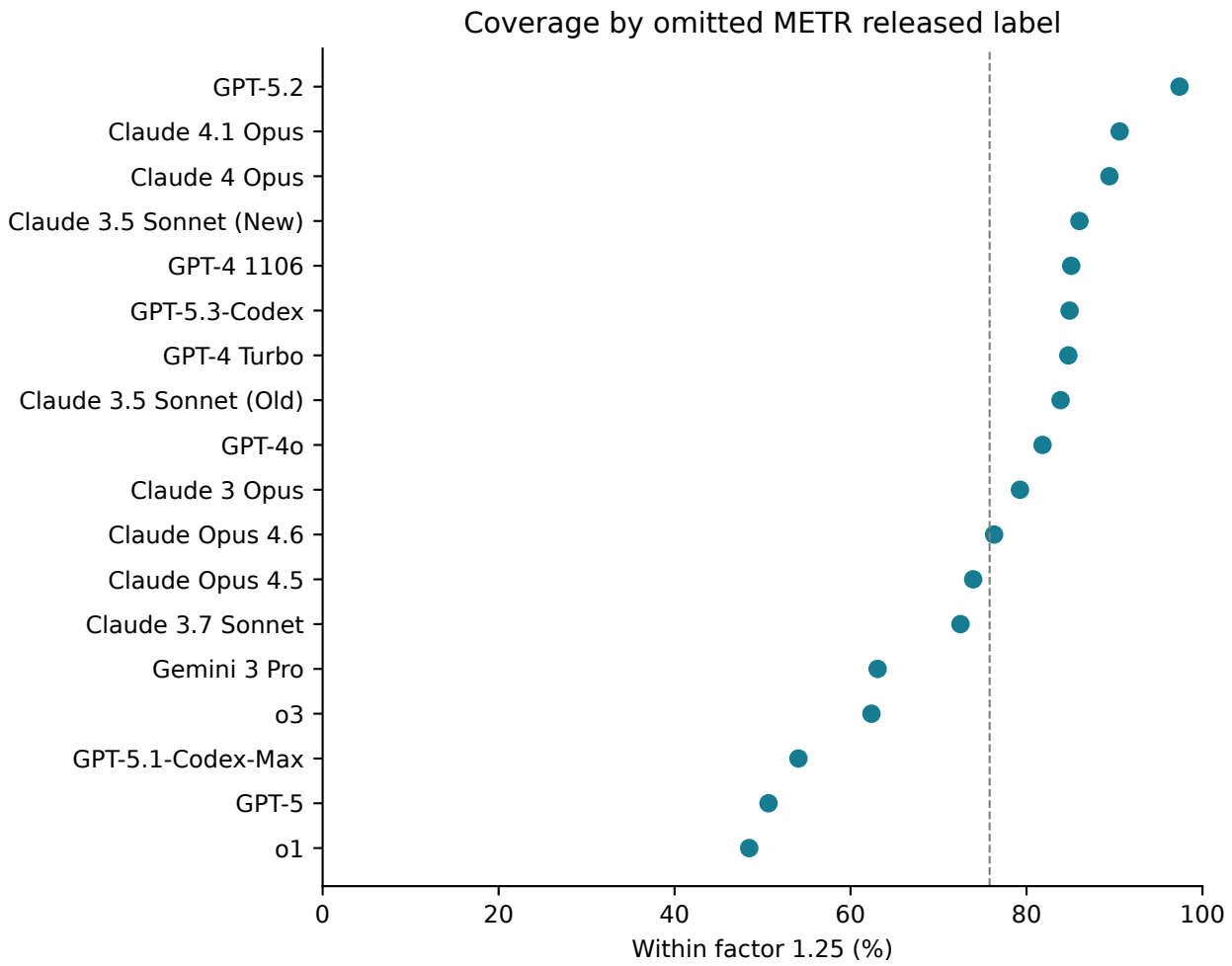


Figure 2: Coverage by omitted METR released label for recorded-token reconstruction. The dashed line marks pooled 75.8% coverage. The factor-1.25 band is descriptive; comparisons share training systems, tasks and anchors. Label identity does not establish immutable returned snapshots.

Table 5: Separate repeat diagnostics, not confidence intervals. Half-repeat results summarize 200 panels from 100 dependent complementary divisions. Constructed results summarize 500 mean-preserving rank-one simulations using cell-specific empirical relative run noise. The range is the 2.5th–97.5th percentile of each diagnostic statistic across panels.

Data construction	Median of median F	Diagnostic range	Median $B_{1.25}$
Full recorded means	1.119	—	75.8%
Half-repeat panels	1.137	1.133–1.142	72.6%
Constructed separable means + run noise	1.046	1.043–1.049	97.1%

identify population costs. Together they show why summary fit, outcome comparability and subgroup support need separate reporting.

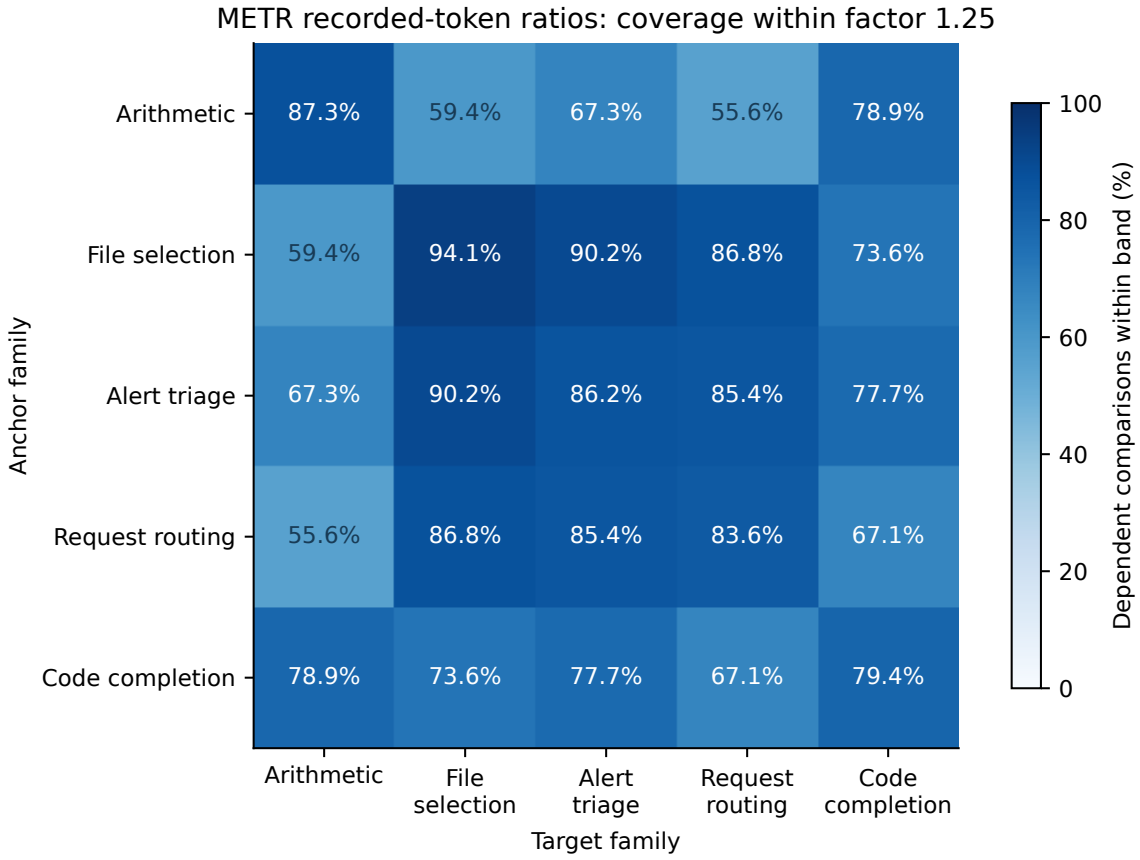


Figure 3: Coverage within factor 1.25 for anchor-domain/target-domain pairs on the same METR matrix. Reversed pairs have equal absolute errors and are not independent. Unequal task counts mean this heatmap and a domain-balanced summary weight comparisons differently from the pooled task-pair statistic.

5.3 Repeat noise and a more flexible comparison

We treat the observed-mean summaries as descriptive. Half-repeat panels test sensitivity to estimating each mean with fewer runs. A fitted rank-one simulation multiplies each separable cell mean by positive episode/observed-cell-mean factors, whose empirical arithmetic mean is one, and averages the original four or six independent resampled factors per cell. It supplies a model-dependent noise-only comparator; it assumes the resampled run noise is an appropriate error model and does not reproduce unknown batch dependence or future instances. It cannot turn a descriptive deviation into a general hypothesis test.

A paired calibration-budget comparison permits the same two anchor means to both a scalar model and a nonnegative affine model with an additive system intercept. Anchors are selected from training systems only; all other task identities are targets. The supplement reports accuracy, positivity constraints and constrained boundary fits, rather than giving the more flexible model free held-target information. Under identical squared-token anchor loss, affine fitting reduces median error from 1.307 to 1.205, but a scalar fitted with squared-log loss achieves 1.162 on the same targets, compared with 1.172 for the affine model fitted with that log loss. Five of 18 affine fits have a zero intercept. Thus fitting loss matters, and these results do not identify a physical overhead mechanism. Both anchors are training-selected extrema; one is the request-routing task containing 31 of the 44 recorded failures. Neither alternative repairs missing outcome or resource evidence.

Table 6: Engineering reconstruction by aggregation. Track and task statistics use different comparison measures. The first two rows retain six routes and elapsed time, but selection changes the tasks; task versus track columns within a row hold suite and resource fixed. Price rows use source schedules, not invoices.

Recorded panel	Routes	Track median F	Track $B_{1.25}$	Task $B_{1.25}$
Elapsed time, selected 22 tasks	6	1.206	59.3%	44.9%
Elapsed time, all 40 tasks	6	1.243	50.7%	25.4%
Output tokens, selected 22 tasks	6	1.170	65.7%	48.8%
Source price estimate, 38 tasks	4	1.139	80.0%	56.0%

6 Missingness and aggregation in engineering records

The auxiliary archive (BRYANN2K, 2026) has 360 records for six requested routes on 40 tasks in ten tracks, including 240 first attempts. Two routes declare the same underlying model. Outcomes are creator-reported; raw outputs, private graders, actual invoices and verifiable returned snapshots are unavailable. All 333 usage-present rows reconcile their native components and source-price arithmetic. The 27 zero-coded unavailable usage rows are retained as missing rather than free inference.

Among first attempts, acceptance is 1/18 with missing usage versus 191/222 with usage. A complete-usage six-route intersection retains 22 tasks in nine tracks. On that subset, elapsed-time track coverage is 59.3%; on all 40 tasks it is 50.7%, and full-suite individual-task coverage is 25.4%. The latter contrast holds tasks and recorded resource fixed while changing aggregation. Restricting to complete token telemetry changes workload too, so its contrast cannot isolate a missingness effect.

No distribution-free, useful upper bound on omitted token expenditure follows from these records: the released turn and wall-time caps do not supply a documented token bound sufficient to bound omitted token expenditure. Zero imputation is unjustified; favorable complete-case fit is not a full-suite result. We therefore use engineering as an audit of selection and aggregation, not as a favorable validation pillar. Full track controls remain in the supplement; Appendix D summarizes the unavailable evidence.

7 What a calibration summary can support

7.1 Task measure and taxonomy

A summary over named categories changes when categories are subdivided unless the comparison measure is carried through the refinement. In a constructed two-system matrix with costs $[1, 1, 100]$ and $[1, 100, 1]$, equal-category pair reconstruction has median factor 100 and zero coverage. Subdividing the stable first family into 100 equal-cost strata gives median 1 and coverage 96.1%, while the specialist discrepancies remain. This is a change of empirical measure, not a calculation bug.

A measure-preserving comparison first fixes probability mass on underlying task pairs, then assigns refined strata their corresponding mass. Distinct children of the same parent require an explicit rule: they must not silently introduce comparison mass absent from the original estimand. The constructed study implements both conventions and recovers the original summary under the fixed pair measure. This correction alone does not make averaging heterogeneous child costs commute with logarithms or calibration. To preserve a general reporting estimand, retain errors at fixed underlying task granularity and aggregate their assigned mass. Deployment mixtures and domain-balanced measures are other choices, not interchangeable estimates of one quantity.

7.2 Accounting consequences require a stronger bound

For a fixed ledger with nonnegative accepted counts n_i, let $W = \sum_i n_i w_i$. If every credited weight satisfies $F^{-1}w_i \le \widehat{w}_i \le Fw_i$, summation gives $F^{-1}W \le \widehat{W} \le FW$. With unchanged positive resource denominators the same bound holds for efficiency. For two positive-work ledgers, their efficiency ratio can be distorted

by up to F^2. A reported efficiency ratio exceeding F^2 certifies its ordering under these assumptions. These bounds concern a *uniform weight error*, which a median pairwise reconstruction error or 75.8% coverage does not supply.

In the constructed ledger, A accepts 1,000 ordinary tasks and B accepts 820 ordinary plus two rare tasks, with equal resource denominators of 100. Weights 1 and 100 give efficiencies 10 and 10.2; changing the rare weight to 80 gives 10 and 9.8. This example illustrates the difference: a small fraction of poorly calibrated comparisons can dominate credited output and reverse an efficiency ordering (Appendix E). This does not measure actual productivity or savings. It explains why a tolerance must follow an accounting requirement rather than be adopted because a convenient pooled error looks small.

7.3 Qualified service costs and sparse overlap

The motivating stronger target is minimum cost among policies meeting a reliability requirement. It differs from the recorded Z and remains unobserved here. It is also a cost per *offered qualified episode*; resources per accepted outcome are cost divided by achieved acceptance under the same policy. A common minimum threshold does not make those quantities coincide. Appendix F gives a counterexample, explicit policy-menu requirements and the unresolved scope of a reference-solve convention.

Sparse overlap adds another limitation. A tree of task bridges always fits its positive labels and has no internal test of invariance. Redundant observations from different systems create tests; longer weak paths accumulate uncertainty. Our constructed graph study illustrates this distinction under a declared noise model, without presenting masked or simulated data as qualified LLM measurements. An empirical sparse qualified-cost experiment is outside the evidence of this paper.

8 Limitations

The archives are convenience samples with related labels, reused tasks, changing dates and unknown policy details. The analyses do not sample a population of systems or future requests. Provider grouping is a proxy; exclusions alter panel composition. Reference, domain, acceptance and alternative-model analyses are retrospective, without confirmatory significance claims. Recorded repeats are sparse, and simulations inherit their stated assumptions. Category counts, anchor availability and aggregation change the summary measure.

The six-grammar arithmetic audit does not certify other graders. Printed-answer correctness and adherence to an output-format contract remain different questions. The engineering release cannot be fully replayed, and source-price reconciliation is not billing verification. Visible response bytes and reported aggregate tokens have different unresolved boundaries; their cross-archive numerical differences cannot identify why fit changes.

The theoretical bounds concern declared positive matrices or weights. They do not establish additive cognition, social value, procurement savings or a validated universal unit. A frozen reference index is a convention whose usefulness needs a separate application study. We have narrowed the contribution accordingly rather than treating incomplete archives as observations of an unmeasured policy frontier.

9 Conclusion

Public evaluation records permit an informative audit of resource-ratio calibration, but favorable aggregate reconstruction is insufficient evidence for a stable conversion between heterogeneous tasks. Independent arithmetic exposes exact-grading failures, and controlled retrospective comparisons expose system, domain, reference, selection and aggregation dependence. A useful calibration report must state its observable, acceptance evidence, task-pair measure and reference convention, and show where subgroup support fails. The released code and records make these bounded findings reproducible; they do not validate a unit of useful inference work.

Reproducibility and data availability

The anonymous supplement includes pinned manifests, normalization and audit code, derived records, generated tables, figures, executed notebooks, and separate verification implementations. The new revision analyses and constructed studies run locally without model calls. The large Reasoning Gym archive and original METR rows are retrieved from their publishers at pinned commits; the supplied downloader verifies their hashes and does not execute downloaded code. The notebook distinguishes recomputed point results from loaded historical resampling ranges. Dated protocols distinguish original retrospective analyses from later amendments. No new inference, paid API evaluations, human-participant study or deployment experiment was performed.

Broader impact

Explicit outcome and resource boundaries can improve the interpretation of model comparisons and prevent unsupported accounting conversions. A task index can also mislead if conventional weights are presented as intrinsic intelligence, business value or employee productivity. Reporting subgroup failures, task mix, uncovered requests and evidence limits reduces this risk. The analyzed panels do not support procurement savings, worker comparisons or production routing claims.

Use of AI assistance

OpenAI Codex assisted with drafting, revision, literature and source inspection, analysis-code development, and automated review. Numerical findings were computed by locally executed programs from pinned public records or explicitly constructed inputs, rather than generated as model measurements. Separate implementations check consequential point results and arithmetic findings. Author review and responsibility for the manuscript, citations, code and conclusions remain essential.

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A Source selection and original sensitivity analyses

Table 7: Reasoning Gym coverage after identity checks and exact cross-system matching. The duplicate column counts original source rows omitted before matching, across all identity-consistent task files; it does not use the matched-instance denominator.

Setting	Systems	Families	Unique instances	Matched records	Source duplicates
Easy	10	88	4,326	43,260	421
Hard	9	86	4,242	38,178	630

The Reasoning Gym manifest records every path, pinned commit, byte size, Git blob identity and SHA-256. Exact matching includes the archived reference, protecting against version differences without certifying its value. Duplicate instances retain their first occurrence. Omitted raw-response transport failures are unavailable. Archive full-credit summaries preserve the strict score-equals-one definition. A stored value of 1.0000000000000002 is a roundoff artifact, not substantive evidence of worse quality; a tolerance of 10^{-12} adds exactly one hard-panel credit (14,226 to 14,227), changing its Gemma 3 27B/spell-backward cell from 17/50 to 18/50. Easy credits, the exploratory high-credit family count and all unconditional byte means are unchanged. The original value and primary rule remain preserved.

Table 8: Post hoc whole-family restriction: every matched first response must be nonempty for every system. Intervals summarize empirical resampling within this restricted dataset. Coupling source positions is a covariance diagnostic; neither interval certifies the full panel or future models.

Setting	Families	Resampling	Median F	2.5–97.5 pct.	$F \leq 1.25$
Easy	19	Independent families	2.089	2.066–2.211	17.0%
		Coupled positions	2.089	2.058–2.216	17.0%
Hard	18	Independent families	2.028	1.963–2.189	18.7%
		Coupled positions	2.028	1.966–2.188	18.7%

In 1,000 original paired-instance byte resamples, 285 easy and 875 hard replicates contain a zero cell mean. We suppress full-panel log-percentile intervals rather than condition on positive survivors. Whole-family zero-free restrictions and coupled source-position resampling were added after inspection. These are descriptive sensitivities, not confidence for the original panel or future tasks.

B Elementary algebra and overlap identification

Let $A_{mi} \in (0, \infty)$ denote any declared matrix entries, recorded or population quantities. For every system, include bridges connecting *all* of its observed task entries, for example every ordered observed pair, with reverse ratios reciprocal. Retain parallel bridges from different systems.

Proposition 1 (Bridge consistency)**.** *If the resulting task graph is connected, the following are equivalent: (i) positive anchored task weights satisfy every bridge $A_{mj}/A_{mi} = w_j/w_i$; (ii) every closed-walk product equals one; (iii) positive factors satisfy $A_{mi} = s_m w_i$ on all observed entries. The weights are unique after fixing one anchor.*

Proof. Ratios of weights telescope, giving (i)⇒(ii). Under (ii), products along paths from the anchor are path-independent and define the unique weights, giving (i). Within-system connectivity makes A_{mi}/w_i the same at every observed task of system m, giving (iii); cancellation gives the reverse implication. A singleton system entry assigns its own factor directly. □

Without within-system bridge closure or connectivity the implication to one factor per system need not hold. A disconnected task graph identifies only component-specific scales.

Proposition 2 (Tree limitation)**.** *One positive independent ratio per edge of a task tree always admits an exact anchored representation.*

Proof. Multiply ratios along the unique path from the anchor to each task. No alternative path imposes a constraint. □

Triangles formed entirely from one system's measured costs telescope automatically and add no independent test of system invariance. Independent comparisons require other systems, redundant paths or held-out entries.

Proposition 3 (Four-cell error certificate)**.** *For systems m, n and tasks i, j, define $D = \log A_{mi} - \log A_{mj} - \log A_{ni} + \log A_{nj}$. Every approximation $\log A_{mi} = \alpha_m + \beta_i + e_{mi}$ has an error of absolute size at least $|D|/4$ in one of these four cells. Its worst multiplicative cell error is at least $\exp(|D|/4)$.*

Proof. The system and task terms cancel, so $D = e_{mi} - e_{mj} - e_{ni} + e_{nj}$ and $|D| \le 4 \max |e|$. □

This is an algebraic statement about supplied means, not a confidence bound on population costs. Selecting the largest observed contrast does not create a prespecified statistical test. The log-additive representation and graph consistency are established methods (Summers, 1973; Jiang et al., 2011).

C METR comparison measures and repeat diagnostics

Table 9: METR-reported aggregate-token reconstruction: 18 system labels on the metadata-selected simple-task panel. Failures retain token usage. Family means weight each included task equally. All-anchor comparisons are dependent and source token components are unavailable.

Aggregation	Rule	Median F	90th pct. F	$F \le 1.25$
66 tasks	Shared scale	1.119	1.424	75.8%
	Constant weights	1.410	2.281	37.8%
	One reference system	1.096	1.510	78.4%
5 families	Shared scale	1.112	1.371	82.8%
	Constant weights	1.597	2.550	24.4%
	One reference system	1.058	1.460	80.6%

Table 10: METR episode-resampling diagnostics: 1,000 independent within-cell resamples; all are finite. Ranges condition on the released task/system panel and do not establish confidence coverage, common reliability or equivalence. The displaced task-level ranges expose nonlinear finite-repeat behavior.

Aggregation	Rule	Observed median F	Resampling 2.5–97.5 pct.
66 tasks	Shared scale	1.119	1.129–1.138
	Constant weights	1.410	1.400–1.417
	One reference system	1.096	1.116–1.137
5 families	Shared scale	1.112	1.098–1.122
	Constant weights	1.597	1.572–1.619
	One reference system	1.058	1.051–1.073

The original within-cell resampling ranges condition on the selected panel and assume independent recorded repeats, without paired model seeds. Their task-level median range lies above the observed median, so the original interval-style figure is not used as an uncertainty figure in the revised main text. The revision instead reports separate perturbation distributions and a noise-only simulation. Neither gives a population confidence interval.

The released table contains 1,829 missing-token rows across other systems and humans. These are not imputed into the metadata-selected complete panel. New system and provider exclusions retain the same 66 tasks and all anchor/target cases unless explicitly stated. Provider-balanced pooling weights each remaining provider group equally before averaging within group. Domain-balanced summaries assign declared mass to domain pairs rather than allow domains with more task identities to dominate. All-observed-success sensitivity is selected using outcomes and has no independent qualification interpretation.

Table 11: A separate two-anchor comparison: 18 omitted labels and 64 withheld targets per fold (1,152 dependent predictions). Training-only minimum/maximum task weights select identical anchors for all methods. Five affine folds hit the zero-intercept boundary. The log-loss scalar outperforms both affine fits on median error, so improved affine fit relative to raw-loss scalar does not identify physical overhead.

Fitting rule	Median F	90th F	$B_{1.25}$
Scalar, squared-token anchor loss	1.307	1.872	39.1%
Nonnegative affine, squared-token loss	1.205	1.861	58.4%
Scalar, squared-log anchor loss	1.162	1.464	69.7%
Nonnegative affine, squared-log loss	1.172	1.468	66.7%

The two-anchor comparison with an affine overhead model uses training-only anchor selection and identical omitted-system information for both models. Constraints and handling of degenerate anchors are part of the released method. Model-dependent accuracy differences are not physical evidence of an overhead mechanism. Detailed per-label, domain-pair and reference results are included as machine-readable tables rather than treated as independent experimental replicates.

D Engineering controls and unavailable evidence

The main source-price panel has four routes and 38 common tasks; the six-route token panel has 22 tasks. Their differences cannot identify a resource effect. Full track-level constant, reference-route and public-description-length controls, together with all task-level contrasts, are retained in the supplementary engineering results. Description-length fits use training routes only. The revised main text reports the controlled within-resource contrasts rather than presenting the restricted source-price fit as validation.

The publisher's private output/grader hashes do not make omitted evidence auditable. Native component reconciliation checks arithmetic on present rows, not completeness of failed-attempt telemetry. Wall time

includes tool and lifecycle effects, and declared turn/time caps do not determine an omitted token count. Frozen prices omit unknown deployment costs and invoices. These limits remain even when a restricted complete-case panel reconstructs closely.

E Constructed taxonomy, accounting and graph studies

Table 12: Constructed taxonomy sensitivity. Only the stable parent is subdivided. Fixed original pair mass conditions on different original parents, preserving the original comparison event as well as the task mass. Keeping masses but changing that event is insufficient.

Stable strata	Pair measure	Median F	$B_{1.25}$
1	Equal current labels	100.000	0.0%
1	Fixed original pair measure	100.000	0.0%
10	Equal current labels	1.000	68.2%
10	Fixed original pair measure	100.000	0.0%
100	Equal current labels	1.000	96.1%
100	Fixed original pair measure	100.000	0.0%

Table 13: Constructed ledgers with equal resource denominator 100. A has 1,000 ordinary acceptances; B has 820 ordinary and two rare acceptances. Reference weights are 1 for ordinary tasks and 100 for the rare task. A small rare-request fraction can control the ordering; these are not measured model efficiencies.

Weight scenario	Efficiency A	Efficiency B	Rare fraction in B
Reference weights	10.00	10.20	0.24%
Rare weight 80 (F=1.25)	10.00	9.80	0.24%
Rare weight 1 (F=100)	10.00	8.22	0.24%

Table 14: Constructed graph noise study: eight tasks, 1,000 replicates, independent Gaussian log-bridge noise with standard deviation 0.12 and no population interaction. Recovery is scored on seven targets of one noiseless separable held-out system with an exact anchor. Analytic root mean squared log error checks the simulation. Different observation counts are explicit; no optimal-design claim follows.

Graph design	Measurements	Simulated log RMSE	Analytic log RMSE
Chain, 7 measurements	7	0.2415	0.2400
Chain + star, 13 measurements	13	0.0840	0.0852
Complete, 28 measurements	28	0.0593	0.0600
Chain, 4 repeats/edge (28)	28	0.1186	0.1200

All values in this appendix come from specified arrays, ledgers or simulated graph observations. They are not API prices, model evaluations or estimates of real productivity. The methods script records seeds, graph edges, weights and comparison measures. Its fixed-measure taxonomy calculation carries parent-pair mass through refinement rather than renormalizing over new category pairs. The accounting example separates a rare heavily weighted task from frequent light tasks. The graph design compares tree-only identification with independently redundant observations under a stated log-noise process. These checks explain the behavior of diagnostics; they cannot validate common-reliability calibration for actual LLMs.

F The motivating qualified-cost proposal and its limits

A task family P_i is a distribution of instances with a declared information, tool, environment, deadline and acceptance contract. A system includes a pinned model and scaffold. An execution policy π specifies budgets,

stopping, retries and verifier feedback. Let $Y \in \{0,1\}$ indicate accepted terminal output, $R^{(r)}$ the complete episode resource including unsuccessful attempts, and $q_{mi\pi} = \mathbb{E}[Y]$ under that distribution. For a finite declared menu Π_{mi}, the motivating cost is

$$C^{(r)}_{mi}(\tau) = \min_{\pi \in \Pi_{mi} : q_{mi\pi} \geq \tau} \mathbb{E}[R^{(r)}_{mi\pi}], \tag{2}$$

with $C^{(r)}_{mi}(\tau) = +\infty$ if the population feasible set is empty. Unresolved empirical qualification remains unknown rather than being declared infeasible. None of the archives measures this target. A selected policy estimate need not equal the menu minimum. Comparable menus require declared admissible budget/retry transformations, independent policy selection and qualification data, and sensitivity to menu expansion. A larger menu can change ratios without changing the outcome contract.

Equation 2 measures resources per offered qualified episode. For a selected minimizing policy with achieved acceptance $q > 0$, long-run resources per accepted output are $\mathbb{E}[R]/q$ under ordinary averaging assumptions. Requiring $q \geq \tau$ does not fix q. In the constructed costs $C_{mi} = 1$ for both systems and tasks, with reliabilities $(0.8, 1)$ and $(1, 0.8)$ at threshold 0.8, offered costs separate exactly. Accepted-output costs are $(1.25, 1)$ and $(1, 1.25)$, whose task ratios disagree by 1.5625. An offered-service or accepted-output calibration is a legitimate declared choice, but they are different estimands.

A reference-solve convention would freeze positive weights $w^{(v)}_i$ with anchor weight one and credit a unique eligible accepted request in family i by $w^{(v)}_i$. For a ledger $\mathcal{L}$,

$$W_v = \sum_{j \in \mathcal{L}} w^{(v)}_{i(j)} Y_j, \qquad \eta^{(r)}_v = \frac{W_v}{\sum_{j \in \mathcal{L}} R^{(r)}_j}. \tag{3}$$

Family assignment must precede observing success, failed episodes retain cost, retries cannot earn duplicate credit, and workflows and their subtasks require explicit allocation. Additivity counts disjoint accepted outcomes by convention; it does not assert cost additivity under caching, batching or reusable artifacts, nor exchangeability of business value. A fixed task mixture can standardize comparisons, but changing or renormalizing uncovered cells changes the question.

A future qualified calibration requires prespecified policy menus, reliability target, task measure, complete resource boundary, independent qualification, a bridge graph with independent redundancy, and held-out prediction under an accounting-based tolerance. The selected error quantile, multiplicity procedure, dependence assumptions and sample size must be justified by operating-characteristic calculations. The single-policy fact that 59/59 independent successes gives a one-sided 95% exact lower bound just above 0.95 is not a whole-study power analysis. We make no executable-study or validation-method claim and do not choose arbitrary planning counts as if they qualified a panel.

G Computational verification and provenance

The supplement maps each empirical claim to pinned inputs and executable code. Separate implementations recompute Reasoning Gym and METR point estimates; exact rational arithmetic checks decimal sums. The revision scripts recompute system/provider/domain/reference diagnostics, quality concentration, repeat perturbations, alternative-model comparisons and explicitly constructed studies. Source arithmetic verification is separate from independent data collection. The historical generator snapshots reveal a compatible precision mechanism but do not replay the original process state, and engineering execution/grading remains unavailable. File hashes protect identity and consistency rather than certify scientific validity.